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#include<stdio.h>
#include"header.h"
int main()
{
    SLL *headptr=0;
    int op,c;
    while(1)
    {
        printf("Enter your choice: \n\033[32m 1)add_begin  2)add_end
        3)add_middle  4)print_node  5)count_node  6)save_file  7)read_file
        8)reverse_print  9)print_rec  10)reverse_rec  11)delete_all
        12)delete_node  13)search_node  14)sort_data  15)reverse_links  16)exit
        \033[0m\n");
        scanf("%d",&op);

        switch(op)
        {
            case 1:addBegin(&headptr);break;
            case 2:addEnd(&headptr);break;
            case 3:addMiddle(&headptr);break;
            case 4:printNode(headptr);break;
            case 5:c = countNode(headptr);
                printf("total count=%d\n",c);break;
            case 6:saveFile(headptr);break;
            case 7:readFile(&headptr);break;
            case 8:reversePrint(headptr);break;
            case 9:printRec(headptr);break;
            case 10:reverseRec(headptr);break;
            case 11:deleteAll(&headptr);break;
            case 12:deleteNode(&headptr);break;
            case 13:searchNode(headptr);break;
            case 14:sortdata(headptr);break;
            case 15:reverseLinks(&headptr);break;
            case 16:exit(0);
            default:printf("\033[31;1;4;5m unknown Choice\033[0m\n");
        }
    }

}

void addBegin(SLL **ptr)
{
    SLL *n=malloc(sizeof(SLL));
    printf("enter rollno,name,marks\n");
    scanf("%d%s%f",&n->rollno,n->name,&n->marks);
    n->next=*ptr;
    *ptr=n;
}

void printNode(SLL *ptr)
{
    if(ptr==0)
    {
        printf("No Record Found\n");
    }
}

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return;
}

printf("\033[34m\n");

while(ptr)
{
printf("%d %s %f\n",ptr->rollno,ptr->name,ptr->marks);
ptr=ptr->next;

}
printf("\033[0m\n");
}

int countNode(SLL *ptr)
{
int c=0;
while(ptr)
{
c++;
ptr=ptr->next;
}
return c;
}

void saveFile(SLL *ptr)
{
if(ptr==0)
{
printf("No record Present\n");
return;
}
printf("\033[34m*****\n");
FILE *fp;
fp=fopen("std.txt","w");

while(ptr)
{
fprintf(fp,"%d %s %f\n",ptr->rollno,ptr->name,ptr->marks);
ptr=ptr->next;
}
printf("Data Saved in file\n");
printf("*****\033[0m\n");
fclose(fp);

}

void reversePrint(SLL *ptr)
{
if(ptr==0)
{
printf("No record Present\n");
return;
}

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}
int c = countNode(ptr);
int i,j;

SLL *t=ptr;

for(i=0;i<c;i++)
{
t=ptr;
for(j=0;j<c-1;j++)
t=t->next;
printf("%d %s %f\n",t->rollno,t->name,t->marks);
}
}

void addEnd(SLL **ptr)
{
SLL *n,*last;
n=malloc(sizeof(SLL));

printf("enter rollno,name & marks\n");
scanf("%d%s%f",&n->rollno,&n->name,&n->marks);
n->next=0;

if(*ptr==0)
*ptr=n;
else
{
last=*ptr;
while(last->next)
last=last->next;
last->next=n;
}
}

void readfile(SLL **ptr)
{
SLL *n,*last;
FILE *fp;
fp=fopen("std.txt","r");
if(fp==0)
{
printf("File Not Present\n");
return ;
}

while(1)
{
n=malloc(sizeof(SLL));

if(fscanf(fp,"%d%s%f",&n->rollno,&n->name,&n->marks)==1)
break;

n->next=0;
if(*ptr==0)

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*ptr=n;
else
{
last=*ptr;
while(last->next)
last=last->next;
last->next=n;
}
}
}
void printRec(SLL *ptr)
{
if(ptr)
{
printf("%d %s %f\n",ptr->rollno,ptr->name,ptr->marks);
if(ptr->next!=0)
printRec(ptr->next);

}
else
printf("No Record Found\n");

}

void reverseRec(SLL *ptr)
{
if(ptr)
{
if(ptr->next!=0)
reverseRec(ptr->next);
printf("%d %s %f\n",ptr->rollno,ptr->name,ptr->marks);
}
else
printf("No Record Found\n");

}

void deleteAll(SLL **ptr)
{
if(*ptr==0)
{
printf("No Record Found\n");
return;
}
SLL *del=*ptr;
int c=1;

while(del)
{
*ptr=del->next;
free(del);
printf("\033[36m Node Deleted:%d\033[0m\n",c++);
sleep(1);
}
}

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del=*ptr;

}
printf("All Nodes Are deleted\n");
}

int searchNode(SLL *ptr)
{
if(ptr==0)
{
printf("No Record found\n");
return;
}
int num;
printf("enter roll no to search\n");
scanf("%d",&num);

while(ptr)
{
if(ptr->rollno==num)
{
printf("%d %s %f\n",ptr->rollno,ptr->name,ptr->marks);
return;
}
ptr=ptr->next;
}
printf("Rollno not found\n");
}

void sortdata(SLL *ptr)
{
if(ptr==0)
{
printf("No record found\n");
return;
}
int c=countNode(ptr);
SLL *p1=ptr,*p2,t;
int i,j;

for(i=0;i<c-1;i++);
{
p2=p1->next;
for(j=0;j<c-1-i;j++)
{
if(strcmp(p1->name,p2->name)>0)
{
t.rollno=p1->rollno;
strcpy(t.name,p1->name);
t.marks=p1->marks;

p1->rollno=p2->rollno;
strcpy(p1->name,p2->name);
p1->marks=p2->marks;

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p2->rollno=t.rollno;
strcpy(p2->name,t.name);
p2->marks=t.marks;
}
p2=p2->next;
}
p1=p1->next;
}
}

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void deleteNode(SLL **ptr)
{
if(*ptr==0)
{
printf("No records found\n");
return;
}

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char name[20];
printf("enter name to delete\n");
scanf("%s",name);

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SLL *prev,*del=*ptr;
while(del)
{
if(strcmp(del->name,name)==0)
{
if(del==*ptr)
*ptr=del->next;
else
prev->next=del->next;

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free(del);
return;
}
prev=del;
del=del->next;
}
printf("Name not found\n");
}

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void reverseLinks(SLL **ptr)
{
if(*ptr==0)
{
printf("No record found\n");
return;
}

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int c=countNode(*ptr);
int i;
if(c>1)

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{
SLL *t=*ptr,**a=malloc(sizeof(SLL *)*c);

while(t)
{
a[i++]=t;
t=t->next;
}
for(i=c-1;i>0;i--)
a[i]->next=a[i-1];

a[0]->next=0;
*ptr=a[c-1];
}
}

void addMiddle(SLL **ptr)
{
SLL *n,*pos;
n=malloc(sizeof(SLL));
printf("enter rollno,name & marks\n");
scanf("%d%s%f",&n->rollno,n->name,&n->marks);

if((*ptr==0)|| (n->rollno < (*ptr)->rollno))
{
n->next=*ptr;
*ptr=n;
}
else
{
pos=*ptr;
while((pos->next!=0)&& (n->rollno)>(pos->next->rollno))
pos=pos->next;

n->next=pos->next;
pos->next=n;
}
}

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